

BROCKEN, Germany

WMO: 104530

Lat: 51.7986N

Lon: 10.6183E

Elev: 1153

StdP: 88.22

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.3	-13.5	-24.9	0.5	-0.7	-20.6	0.7	-9.0	32.5	-0.2	29.9	-1.2	12.1	0	1.208

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	20.4	14.5	18.9	13.8	17.3	13.0	15.5	18.8	14.5	17.6	13.6	16.5	7.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	11.7	16.9	13.2	10.9	15.7	12.3	10.3	14.9	47.6	19.1	44.6	17.6	41.9	16.5	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.8	25.7	23.1	DB	-16.5	23.8	5.2	1.8	-20.2	25.1	-23.2	26.1	-26.1	27.1	-29.8	28.4	(4)
(5)			WB	-16.3	17.6	4.9	1.3	-19.9	18.6	-22.8	19.4	-25.5	20.1	-29.1	21.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-4.1	-5.0	-1.7	1.0	6.1	8.8	11.2	11.3	7.8	4.7	0.0	-1.8	(6)	
	DBStd	6.98	4.72	5.15	4.25	4.17	4.27	4.14	3.95	3.76	3.55	3.97	4.34	3.68	(7)	
	HDD10.0	2684	438	419	363	270	134	73	34	29	86	172	299	366	(8)	
	HDD18.3	5509	696	652	622	519	378	287	223	219	315	424	549	624	(9)	
	CDD10.0	219	0	0	0	1	14	36	71	71	21	7	1	0	(10)	
	CDD18.3	3	0	0	0	0	0	1	1	2	0	0	0	0	(11)	
	CDH23.3	6	0	0	0	0	0	0	1	4	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	11.3	15.8	13.0	11.7	9.5	8.5	8.7	8.8	9.0	10.8	12.6	13.1	14.4	(14)	
(15) Precipitation	PrecAvg	1722	189	139	143	106	127	119	143	129	137	154	171	203	(15)	
	PrecMax	2725	439	330	360	304	317	261	428	278	508	451	342	515	(16)	
	PrecMin	1080	40	27	33	23	38	46	12	34	43	36	13	35	(17)	
	PrecStd	343	91	71	77	58	60	43	80	56	85	84	78	101	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	8.5	11.6	13.2	17.6	21.3	22.2	23.4	18.2	16.4	11.9	8.3	(19)	
		MCWB	2.1	2.6	4.9	7.6	11.1	13.8	15.8	15.8	13.0	9.9	6.8	2.7	(20)	
	2%	DB	4.0	5.9	8.6	10.8	15.7	18.9	20.2	20.4	16.6	14.2	9.8	6.5	(21)	
		MCWB	2.1	2.5	3.7	6.7	10.2	13.7	14.9	14.4	12.3	9.6	5.5	1.8	(22)	
	5%	DB	2.8	3.9	5.9	8.9	14.2	16.9	18.9	18.5	15.1	12.1	8.0	5.0	(23)	
		MCWB	1.6	1.4	3.6	5.6	9.5	12.7	14.0	13.7	11.6	8.8	4.6	2.3	(24)	
	10%	DB	1.1	2.0	3.9	7.3	12.6	15.0	17.3	16.8	13.2	10.3	6.3	3.7	(25)	
		MCWB	-0.1	0.6	2.8	4.7	8.7	11.9	13.2	12.9	10.9	8.0	4.5	1.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	5.6	6.2	8.5	12.2	16.1	17.2	17.0	14.5	12.2	7.8	5.5	(27)	
		MCDB	4.4	6.5	9.2	11.5	15.4	19.0	20.6	22.2	16.6	14.0	9.3	6.2	(28)	
	2%	WB	3.1	3.6	4.9	7.3	11.1	14.3	15.8	15.2	13.2	11.1	6.9	4.5	(29)	
		MCDB	3.2	4.6	6.8	9.8	14.5	17.3	19.0	18.8	15.2	13.4	8.3	5.1	(30)	
	5%	WB	1.7	1.7	3.9	6.3	10.1	13.1	14.7	14.3	12.3	9.6	5.9	3.2	(31)	
		MCDB	2.3	3.0	5.6	8.5	13.4	16.3	17.8	17.4	14.0	11.1	7.3	4.1	(32)	
	10%	WB	0.2	0.6	2.7	5.0	9.1	12.1	13.7	13.4	11.1	8.5	4.8	2.1	(33)	
		MCDB	0.6	1.6	4.0	7.0	12.0	14.7	16.7	16.2	12.7	9.6	5.9	2.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	4.0	3.8	4.5	5.0	4.9	5.3	5.0	4.2	4.1	3.7	3.7	(35)	
		MCDBR	5.5	5.5	5.6	6.2	6.4	6.6	6.7	7.1	6.6	5.8	5.7	5.8	(36)	
	5% WB	MCWBR	4.9	4.2	4.0	4.1	3.9	4.6	4.7	4.1	4.6	4.1	4.2	4.5	(37)	
		MCWBR	4.9	5.0	5.1	6.0	6.3	6.2	6.6	6.4	5.8	5.1	4.8	4.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.268	0.286	0.316	0.344	0.347	0.345	0.358	0.352	0.325	0.304	0.287	0.264	(40)	
		taud	2.468	2.459	2.410	2.369	2.373	2.386	2.346	2.383	2.465	2.522	2.509	2.459	(41)	
	Ebn at Noon		771	853	885	894	905	907	887	875	865	820	739	714	(42)	
		Edh at Noon	54	72	92	109	114	113	116	106	87	68	52	47	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.70	1.40	2.48	4.00	4.91	5.29	4.97	4.24	3.08	1.77	0.81	0.54	(44)	
	RadStd		0.08	0.19	0.28	0.48	0.57	0.43	0.67	0.39	0.38	0.19	0.10	0.06	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page